

Homework #5

Due: 11:59pm, March 2

Extra Exercises: OS 8.28 & 8.291. **[OS 8.49]**

Suppose $x_1[n]$ is an infinite-length, stable (i.e. absolutely summable) sequence with z -transform given by

$$X_1(z) = \frac{1}{1 - \frac{1}{3}z^{-1}}.$$

Suppose $x_2[n]$ is a finite-length sequence of length N , and the N -point DFT of $x_2[n]$ is

$$X_2[k] = X_1(z)|_{z=e^{j2\pi k/N}}, \quad k = 0, 1, \dots, N-1.$$

Determine $x_2[n]$.

2. **[OS 8.55]**

Let $X[k]$ denote the N -point DFT of the N -point sequence $x[n]$.

(a) Show that if

$$x[n] = -x[N-1-n],$$

then $X[0] = 0$. Consider separately the cases of N even and N odd.

(b) Show that if N is even and if

$$x[n] = x[N-1-n],$$

then $X[N/2] = 0$.

3. **[OS 8.64: Approximate inverse filter]**

A problem that often arises in practice is one in which a distorted signal $y[n]$ is the output that results when a desired signal $x[n]$ has been filtered by an LTI system. We wish to recover the original signal $x[n]$ by processing $y[n]$. In theory, $x[n]$ can be recovered from $y[n]$ by passing $y[n]$ through an inverse filter having a system function equal to the reciprocal of the system function of the distortion filter.

Suppose that the distortion is caused by an FIR filter with impulse response

$$h[n] = \delta[n] - \frac{1}{2}\delta[n - n_0],$$

where n_0 is a positive integer, i.e. the distortion of $x[n]$ takes the form of an echo at delay n_0 .

(a) Determine the z -transform $H(z)$ and the N -point DFT $H[k]$ of the impulse response $h[n]$. Assume that $N = 4n_0$.

- (b) Let $H_i(z)$ denote the system function of the inverse filter, and $h_i[n]$ be the corresponding impulse response. Determine $h_i[n]$. Is this an FIR or an IIR filter? What is the duration of $h_i[n]$?
- (c) Suppose that we use an FIR filter of length N in an attempt to implement the inverse filter, and let the N -point DFT of the FIR filter be

$$G[k] = 1/H[k], \quad k = 0, 1, \dots, N-1.$$

What is the impulse response $g[n]$ of the FIR filter? Hint: Use the fact

$$\mathcal{F}_{\text{DFT}} \left\{ \sum_{l=-\infty}^{\infty} \delta[n - lN] \right\} = \frac{2\pi}{N} \sum_{m=-\infty}^{\infty} \sum_{k=0}^{N-1} \delta \left(\omega - 2\pi m - \frac{2\pi k}{N} \right)$$

and $g[n]$ for $n = 0, 1, \dots, N-1$ is computed by

$$\begin{aligned} g[n] &= \frac{1}{N} \sum_{k=0}^{N-1} G[k] e^{j2\pi kn/N} \\ &= \frac{1}{2\pi} \int_{-\epsilon}^{2\pi-\epsilon} \frac{2\pi}{N} \sum_{m=-\infty}^{\infty} \sum_{k=0}^{N-1} \delta \left(\omega - 2\pi m - \frac{2\pi k}{N} \right) H_i(e^{j\omega}) e^{j\omega kn} d\omega \\ &= \mathcal{F}_{\text{DFT}}^{-1} \left\{ \frac{2\pi}{N} \sum_{m=-\infty}^{\infty} \sum_{k=0}^{N-1} \delta \left(\omega - 2\pi m - \frac{2\pi k}{N} \right) H_i(e^{j\omega}) \right\}, \end{aligned}$$

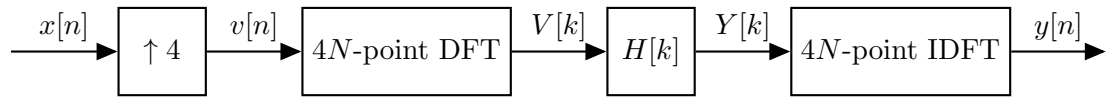
where $\epsilon > 0$ is a small constant. Then you can express $g[n]$ in the interval $\{0, 1, \dots, N-1\}$ as a time-aliased version of $h_i[n]$. Note that $g[n]$ has a nonzero value only at $n = 0, n_0, 2n_0, 3n_0$. Simplify the expression.

- (d) Perform the convolution of $g[n]$ and $h[n]$, and this determine how well the FIR filter with N -point DFT $G[k] = 1/H[k]$ implements the inverse filter. That is, compare $g[n] * h[n]$ to $\delta[n]$. Can we decrease the error if we set N to be n_0 multiplied by a larger positive integer than 4 (e.g. $N = 8n_0$)?

4. [OS 8.67]

In this problem, you will examine the use of the DFT to implement the filtering necessary for the discrete-time interpolation, or upsampling, of a signal. Assume that the discrete-time signal $x[n]$ was obtained by sampling a continuous-time signal $x_c(t)$ with a sampling period T . Moreover, the continuous-time signal is appropriately bandlimited: i.e., $X_c(j\Omega) = 0$ for $|\Omega| \geq 2\pi/T$. For this problem, we will assume that $x[n]$ has length N . It is not strictly possible to have a signal that is both perfectly bandlimited and of finite duration, but this is often assumed in practical systems processing finite-length signals which have very little energy outside the band $|\Omega| \leq 2\pi/T$.

We wish to implement a 1:4 interpolation, i.e. increase the sampling rate by a factor of 4. We can perform this sampling rate conversion using a sampling rate expander followed by an appropriate lowpass filter. For this problem, assume that the filter has an impulse response $h[n]$ that is $N+1$ points long. The following figure depicts such a system, where $H[k]$ is the $4N$ -point DFT of the impulse response of the lowpass filter. Note that both $v[n]$ and $y[n]$ are $4N$ -point sequences.



Specify the DFT $H[k]$ such that this system implements the desired upsampling system. Think carefully about the phase of the values of $H[k]$. The impulse response $h[n]$ is supported on $0, 1, \dots, N$.